

CHAPTER 13

WAVES

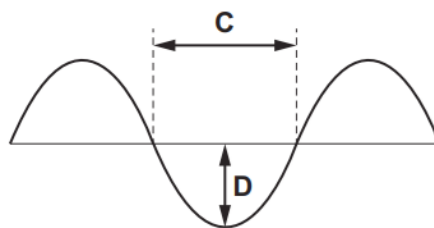
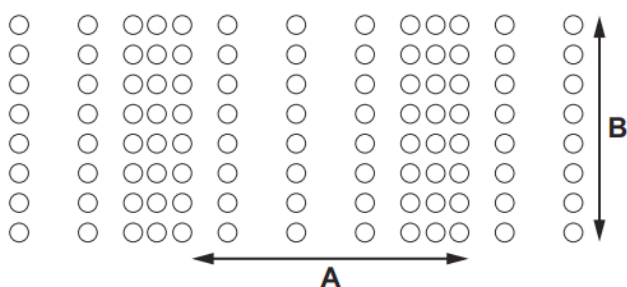
3.1 General properties of waves

- 1 Know that waves transfer energy without transferring matter
- 2 Describe what is meant by wave motion as illustrated by vibrations in ropes and springs and by experiments using water waves
- 3 Describe the features of a wave in terms of wavefront, wavelength, frequency, crest (peak), trough, amplitude and wave speed
- 4 Define the terms:
 - (a) frequency as the number of wavelengths that pass a point per unit time
 - (b) wavelength as the distance between two consecutive, identical points such as two consecutive crests
 - (c) amplitude as the maximum distance from the mean position
- 5 Recall and use the equation
wave speed = frequency $\times$ wavelength
 $v = f\lambda$
- 6 Know that for a transverse wave, the direction of vibration is at right angles to the direction of the energy transfer, and give examples such as electromagnetic radiation, waves on the surface of water, and seismic S-waves (secondary)
- 7 Know that for a longitudinal wave, the direction of vibration is parallel to the direction of the energy transfer, and give examples such as sound waves and seismic P-waves (primary)
- 8 Describe how waves can undergo:
 - (a) reflection at a plane surface
 - (b) refraction due to a change of speed
 - (c) diffraction through a gap
- 9 Describe how wavelength and gap size affects diffraction through a gap
- 10 Describe the use of a ripple tank to show:
 - (a) reflection at a plane surface
 - (b) refraction due to a change in speed caused by a change in depth
 - (c) diffraction due to a gap
 - (d) diffraction due to an edge
- 11 Describe how wavelength affects diffraction at an edge

1. O/N/23/12/Q24

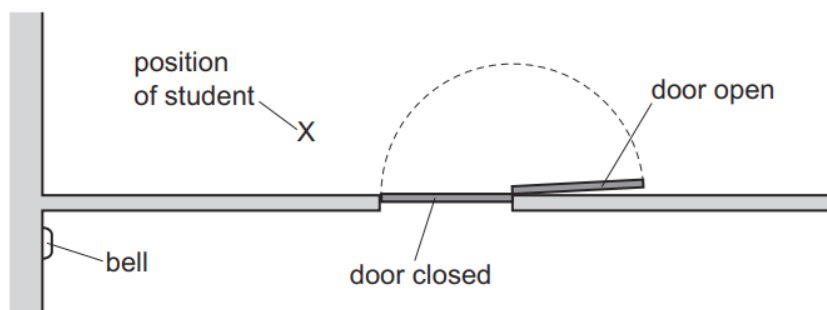
The diagram shows two waveforms.

Which arrow shows the wavelength of a longitudinal wave?



2. O/N/23/12/Q25

A bell, on a wall outside a classroom, rings. A student is standing at point X shown on the view from above.



The door of the classroom is closed and the student in the room at X hears the bell ring.

The door is opened, and the sound heard by the student from the bell is louder.

What causes the sound reaching the student to be louder?

- A** diffraction
- B** dispersion
- C** reflection
- D** refraction

3. O/N/22/11/Q23

Which statement is correct?

- A** Both longitudinal waves and transverse waves can travel at the speed of light.
- B** Both longitudinal waves and transverse waves of similar wavelengths must have similar frequencies.
- C** Both longitudinal waves and transverse waves can travel in a vacuum.
- D** Both longitudinal waves and transverse waves can travel in water.

4. M/J/22/11/Q19

A wave in the sea collides with a cliff.

A crest of the wave hits the cliff once every 6.0 s.

The horizontal distance between a crest and the adjacent trough of the wave is 4.5 m.

What is the speed of the wave?

- A** 0.67 m/s **B** 0.75 m/s **C** 1.3 m/s **D** 1.5 m/s

5. M/J/22/11/Q20

A sound wave refracts as it passes from air to water.

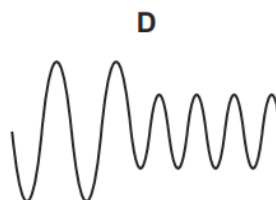
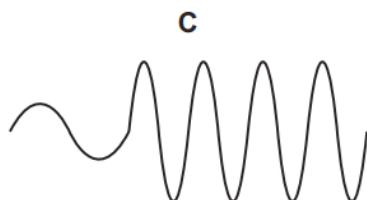
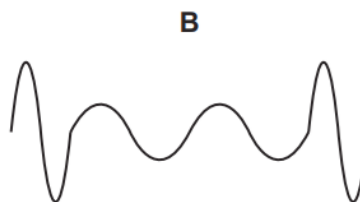
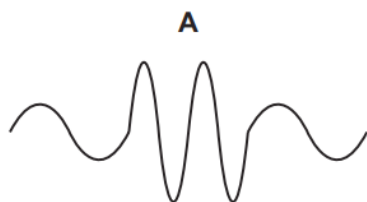
Which quantities change?

- A** speed and frequency only
B speed and wavelength only
C wavelength and frequency only
D speed, frequency and wavelength

6. O/N/21/11/Q17

A water wave on a pond passes into a region where the wave travels more slowly. It then returns to a region where it travels at the original speed. Assume that the frequency of the wave stays constant.

Which side-view of the wave is correct?



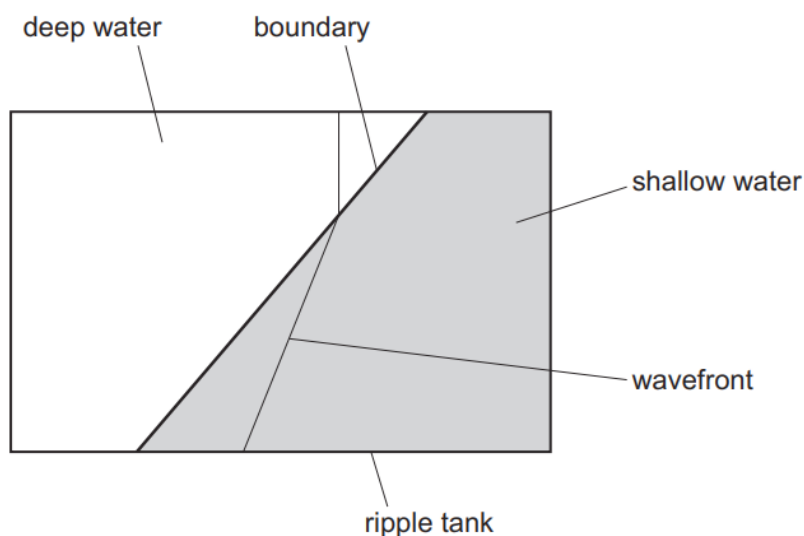
7. O/N/21/11/Q18

What is meant by the *amplitude* of a wave?

- A** the distance between the extreme points of a particle's motion
- B** the maximum distance a particle moves from its equilibrium position
- C** the maximum energy carried by the wave
- D** the maximum power of the vibrations carrying the wave

8. O/N/21/11/Q19

The diagram shows one wavefront of a wave as it travels from deep water to shallow water in a ripple tank.



What happens as the wavefront moves into the shallow water?

- A** The speed of the wavefront increases.
- B** The speed of the wavefront decreases.
- C** The wavelength of the wave remains constant.
- D** The wavelength of the wave increases.

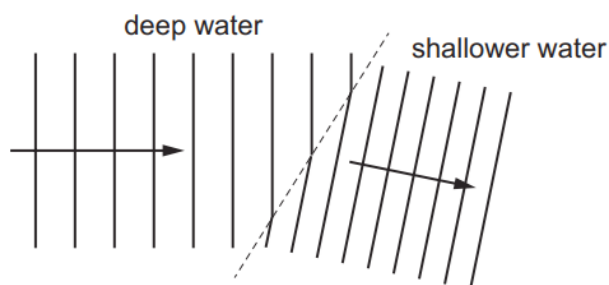
9. O/N/21/12/Q22

Which example describes a transverse wave being refracted?

- A** infrared waves travelling in an optical fibre
- B** light waves from an object entering a magnifying glass
- C** ultrasound waves being used to clean jewellery
- D** water waves hitting a barrier

10. M/J/21/11/Q20

A water wave in a ripple tank refracts as it moves from deep water into shallower water.



What happens to the speed and to the frequency of the wave as it moves into shallower water?

	speed	frequency
A	decreases	stays constant
B	increases	stays constant
C	stays constant	decreases
D	stays constant	increases

11. O/N/20/11/Q26

An electromagnetic wave has a speed of 3.0×10^8 m/s and a wavelength of 10 cm.

What is the frequency of the wave?

- A** 3.3×10^{-10} Hz
- B** 3.3×10^{-8} Hz
- C** 3.0×10^7 Hz
- D** 3.0×10^9 Hz

12. M/J/20/12/Q30

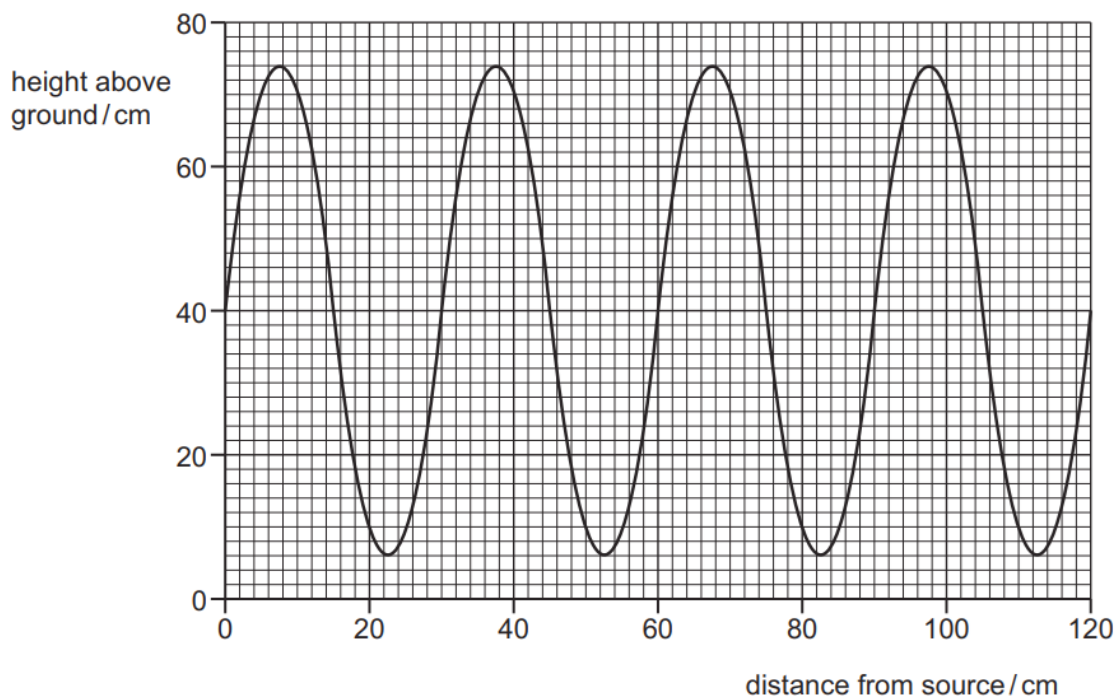
A vibrator produces 12 wavelengths on the surface of water in 10 s. The spacing between the first crest and the third crest is 60 cm.

What is the speed of the wave?

- A** 24 cm/s **B** 25 cm/s **C** 36 cm/s **D** 72 cm/s

13. M/J/20/12/Q29

The diagram shows how the height above the ground of a rope transmitting a wave varies with distance from the source of vibrations at one instant in time.



What is the amplitude of the wave shown?

- A** 30 cm **B** 34 cm **C** 72 cm **D** 74 cm

14. O/N/19/11/Q25

The frequency of microwaves used in an experiment is 3.0×10^9 Hz. Their speed in air is 3.0×10^8 m/s.

What is a possible frequency and a possible speed in air of a radio wave?

	frequency / Hz	<u>speed in air</u> m/s
A	3.0×10^5	1500
B	3.0×10^5	3.0×10^8
C	3.0×10^{14}	1500
D	3.0×10^{14}	3.0×10^8

15. O/N/19/12/Q22

Which description defines the amplitude of a water wave in a ripple tank?

- A** half of the vertical distance between a trough and a peak
- B** the horizontal distance between adjacent peaks
- C** the horizontal distance between a trough and an adjacent peak
- D** the vertical distance between a trough and a peak

16. O/N/19/12/Q23

Here are four statements about wavefronts.

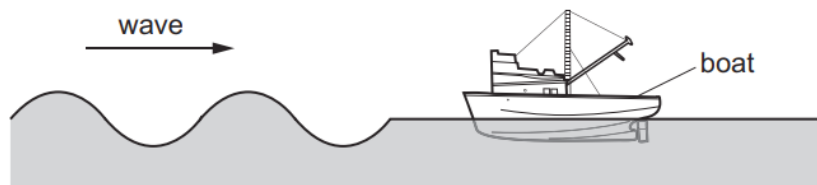
- 1 Wavefronts from a point source are straight lines.
- 2 Wavefronts from a point source are circular.
- 3 Wavefronts are parallel to the direction of energy movement.
- 4 Wavefronts are perpendicular to the direction of energy movement.

Which two statements are correct?

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

17. M/J/19/12/Q25

A wave on water travels towards a stationary boat.



As the wave passes, the boat starts to move up and down and gains kinetic energy from the wave.

Which property of the wave changes as the wave passes the boat?

- A** amplitude
- B** frequency
- C** speed
- D** wavelength

18. M/J/19/12/Q26

Which waves are longitudinal?

- A** gamma rays
- B** light waves
- C** ultrasound waves
- D** X-rays

19. O/N/18/11/Q18

Which term is the number of crests of a wave passing a point per second?

- A** amplitude
- B** frequency
- C** speed
- D** wavelength

20. O/N/18/12/Q22

Which statement about a water wave is correct?

- A** The amplitude is the vertical distance between a trough and a peak.
- B** The frequency is the number of troughs passing a point in one second added to the number of peaks passing a point in one second.
- C** The speed is the horizontal distance travelled per second by a peak.
- D** The wavelength is the horizontal distance between a trough and a peak.

21. M/J/18/12/Q26

A communications system uses microwaves with a wavelength of 15 cm. The speed of light is 3.0×10^8 m/s.

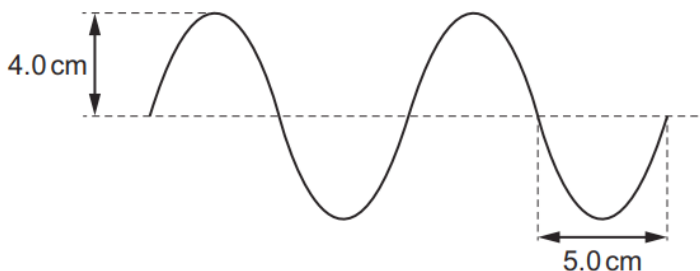
What is the frequency of the microwaves?

- A** 2.0×10^7 Hz **B** 4.5×10^7 Hz **C** 2.0×10^9 Hz **D** 4.5×10^9 Hz

22. O/N/17/11/Q18

A wave is travelling in water in a glass-sided tank.

The diagram shows a side view of the wave at one instant.



What are the amplitude and the wavelength of the wave?

	amplitude / cm	wavelength / cm
A	4.0	5.0
B	4.0	10
C	8.0	5.0
D	8.0	10

23. O/N/17/11/Q19

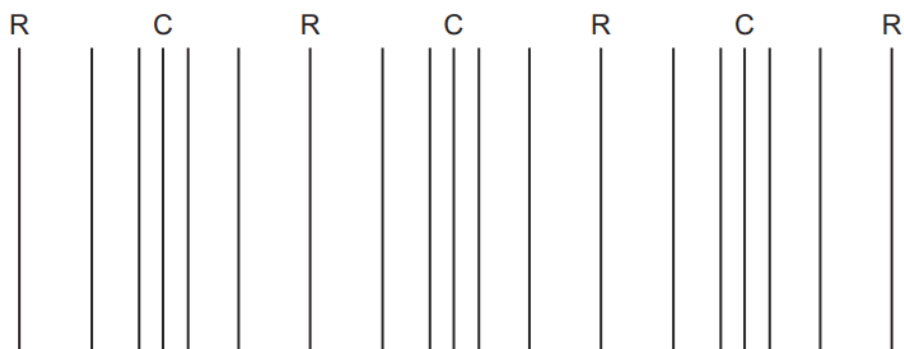
An earthquake wave travels through the solid surface of the Earth from east to west. The solid surface vibrates in a north-south direction.

How can the earthquake wave be described?

- A** electromagnetic
- B** longitudinal
- C** sound
- D** transverse

24. O/N/17/12/Q19

A sound wave in a solid is represented by a series of lines. The diagram shows compressions C and rarefactions R at one instant in time.



Which statement is correct?

- A** All particles on one line move in the same direction at the same time.
- B** The direction of travel of the sound wave is parallel to the lines.
- C** The distance between a compression and its adjacent rarefaction is a wavelength.
- D** The pattern of lines represents a transverse wave.

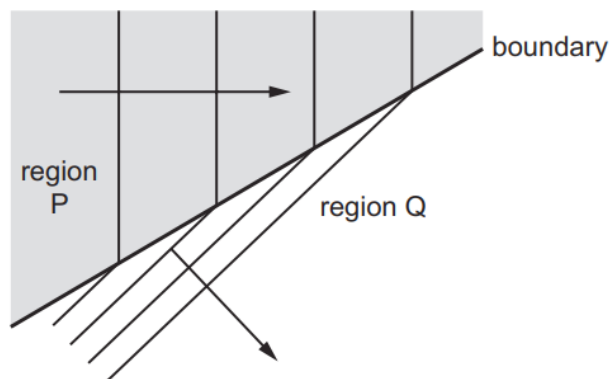
25. M/J/17/11/Q26

Which waves are longitudinal?

- A** sound waves in water
- B** ultra-violet waves in air
- C** waves on the surface of water
- D** X-rays in a vacuum

26. M/J/17/12/Q26

The diagram shows a water wave in a ripple tank being refracted at the boundary between region P and region Q.

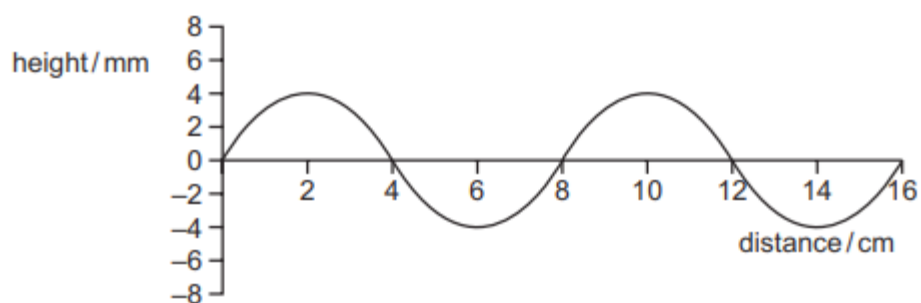


How does the wave in region P compare with the wave in region Q?

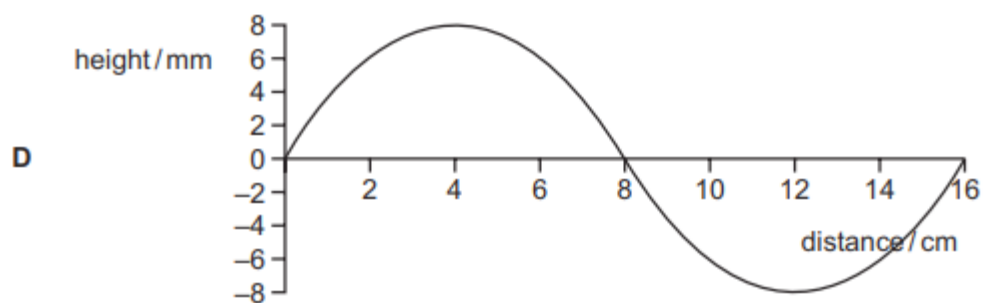
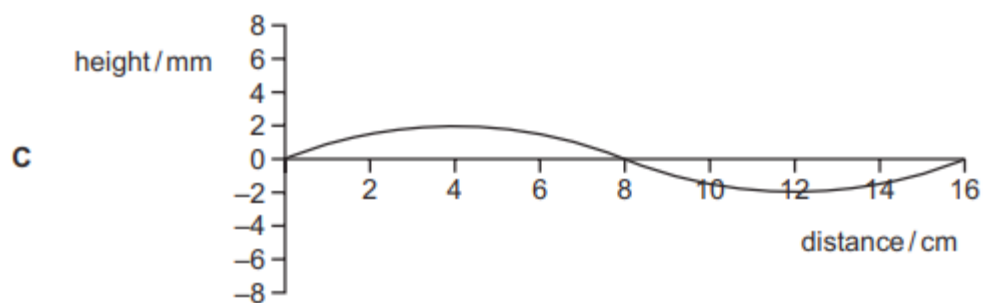
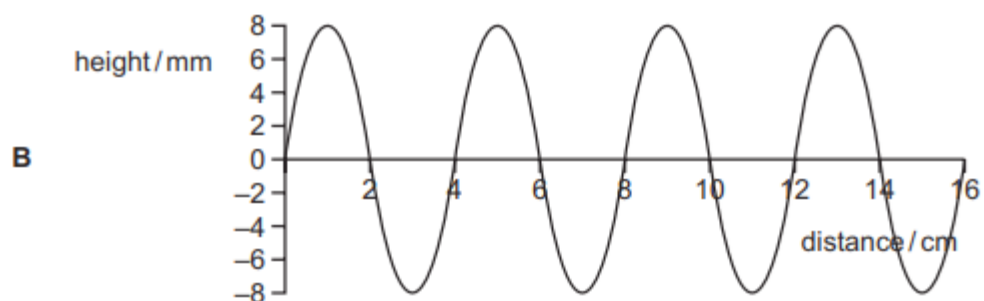
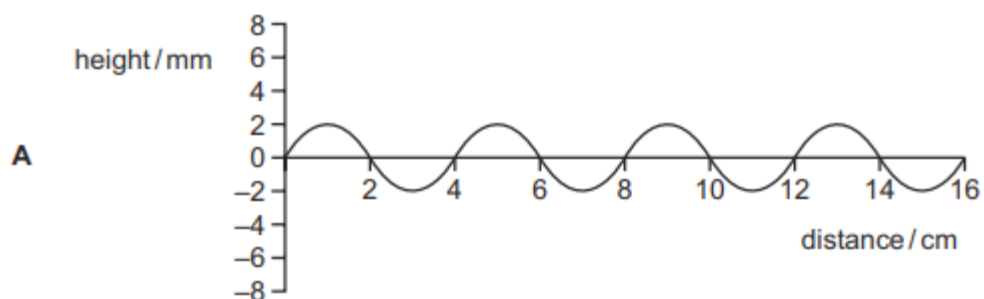
	wavelength of wave in region P	speed of wave in region P
A	larger	faster
B	larger	slower
C	smaller	faster
D	smaller	slower

27. M/J/17/12/Q27

The graph shows how the height of a water wave varies with distance along the wave.



Which graph shows a wave with twice the amplitude, half the frequency, and the same speed?



28. O/N/16/11/Q21

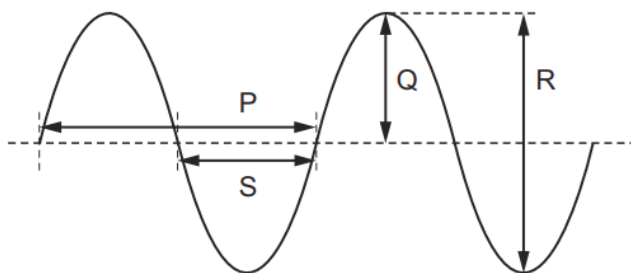
A phone rings and lights up.

Which types of wave are associated with the phone's operation?

	light	sound
A	longitudinal	longitudinal
B	longitudinal	transverse
C	transverse	longitudinal
D	transverse	transverse

29. O/N/16/11/Q22

The diagram shows the surface of water as a wave passes across a ripple tank.



Which lengths represent the amplitude and the wavelength of the wave?

	amplitude	wavelength
A	Q	P
B	Q	S
C	R	P
D	R	S

30. O/N/16/12/Q25

Which statement about waves is correct?

- A** All electromagnetic waves are transverse waves.
- B** All transverse waves travel at 300 000 000 m/s.
- C** Longitudinal waves cannot be refracted.
- D** Water waves are longitudinal waves.

31. M/J/16/12/Q17

A water wave in a ripple tank refracts as it passes from deep water to shallow water.

Which properties change as the wave refracts?

- A frequency and amplitude
- B frequency and wavelength
- C speed and frequency
- D speed and wavelength

32. M/J/16/11/Q19

For a transverse wave, what is a *wavefront*?

- A a line joining all points on the same crest of a wave
- B a line showing the displacement of a wave
- C the energy content of a wave
- D the first part of a wave to reach a point

33. M/J/16/11/Q20

A longitudinal wave passes along a spring. The coils of the spring vibrate from side to side.

The diagram shows the positions of the coils at one particular time.

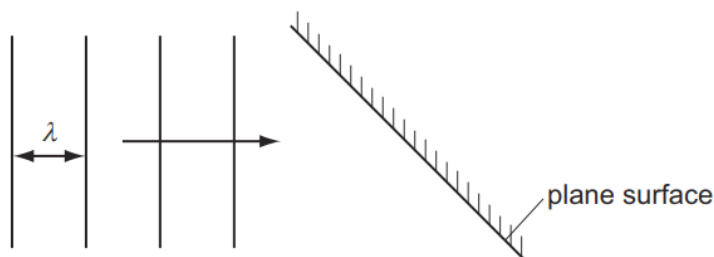


Which positions are one wavelength apart?

- A W and X
- B W and Z
- C X and Z
- D Y and Z

34. O/N/15/11/Q19

In an experiment using a ripple tank, plane wavefronts arrive at a plane surface.



Which row correctly describes the waves after they are reflected from the surface?

	speed of waves	wavelength λ
A	larger	shorter
B	smaller	shorter
C	the same	longer
D	the same	the same

35. O/N/15/12/Q21

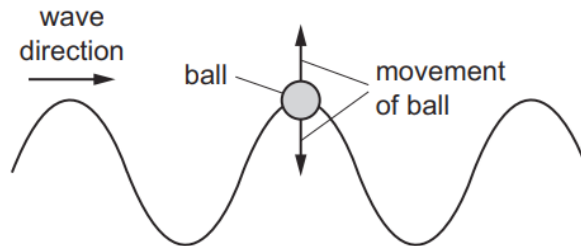
An echo sounder produces ultrasound of frequency 24 kHz. The ultrasound travels in water at a speed of 1.5 km/s.

What is the wavelength in water of ultrasound of this frequency?

- A** 0.063 m **B** 16 m **C** 36 m **D** 63 m

36. M/J/15/12/Q24

A ball floating in a ripple tank begins to move vertically up and down as a wave passes beneath it. The ball does not move horizontally.



Which statement is correct?

- A** Both energy and water are transferred in the wave direction.
B Energy is not transferred in the wave direction but water is.
C Energy is transferred in the wave direction but water is not.
D Neither energy nor water is transferred in the wave direction.

37. O/N/14/11/Q19

A wave of frequency 13 000 Hz travels 1300 m in 4.0 s.

What is the wavelength of the wave?

- A** 0.025 m **B** 0.40 m **C** 2.5 m **D** 40 m

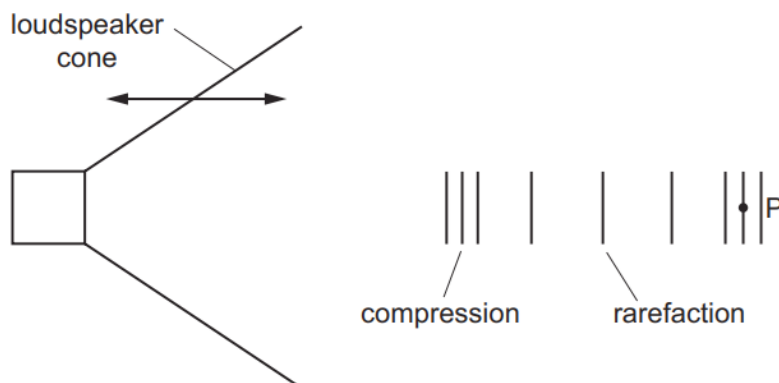
38. M/J/14/12/Q25

What is the frequency of a wave?

- A** The number of waves passing a fixed point per second.
B The number of peaks added to the number of troughs passing a fixed point per second.
C The time taken for one wave to pass a fixed point.
D The time taken for the displacement to change from maximum to minimum.

39. M/J/14/11/Q22

Compressions and rarefactions are sent out from a loudspeaker cone as it vibrates backwards and forwards. The frequency of vibration is 50 Hz.



A compression is at point P. How much time elapses before the next rarefaction arrives at P?

- A** 0.010 s **B** 0.020 s **C** 25 s **D** 50 s

40. M/J/14/11/Q23

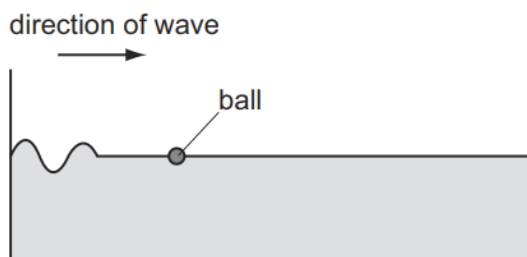
Water waves refract at a boundary between deep water and shallow water.

What is the effect on the frequency, wavelength and speed of the waves at the boundary?

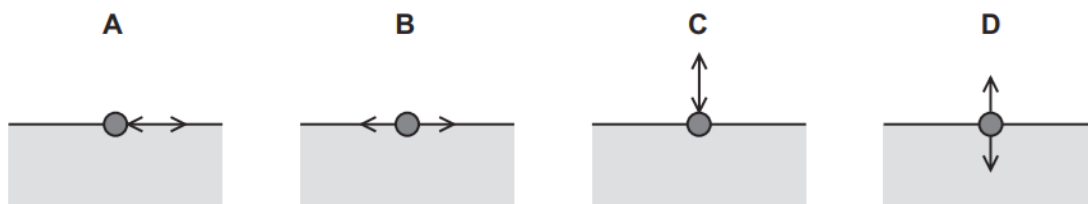
	frequency	wavelength	speed
A	changes	changes	stays the same
B	changes	stays the same	stays the same
C	stays the same	changes	changes
D	stays the same	stays the same	changes

41. O/N/13/11/Q19

The diagram shows a ball floating in a tank of water.

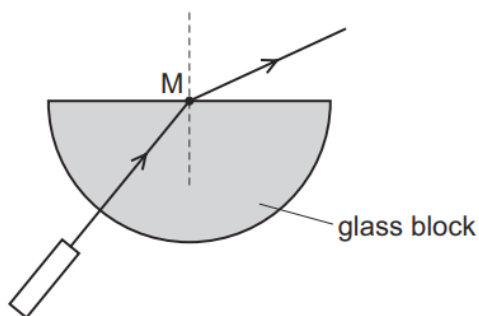


Which diagram shows the movement of the ball as the wave passes?



42. O/N/13/12/Q23

A ray of red light from a laser passes into a semi-circular glass block.



What is shown at M?

- A** dispersion
- B** rarefaction
- C** reflection
- D** refraction

43. M/J/13/12/Q22

Energy can be transferred in many different ways.

In which situation is energy transferred by wave motion?

- A** colliding atoms in a heated copper rod
- B** fast-moving electrons in a cathode-ray oscilloscope
- C** hot water rising in a heated saucepan
- D** ripples passing across water in a ripple tank

44. M/J/13/12/Q23

A wave has a frequency of 10 kHz.

Which pair of values of its speed and wavelength is possible?

	<u>speed</u> m/s	<u>wavelength</u> m
A	330	0.33
B	330	33
C	3.0×10^8	30
D	3.0×10^8	3.0×10^4

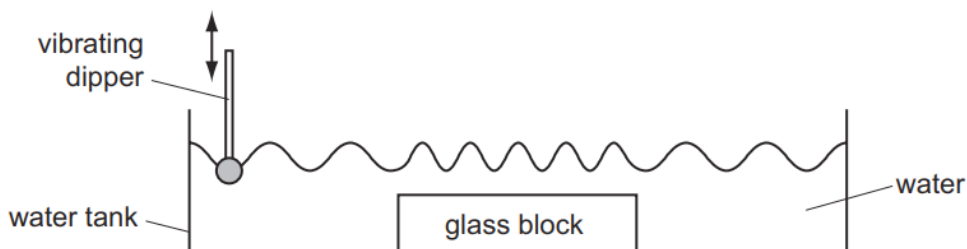
45. M/J/13/12/Q24

Which of the following travels as a longitudinal wave?

- A** a radio wave in air
- B** a sound wave in a solid
- C** a wave on a rope shaken from side to side
- D** an infra-red wave in space

46. M/J/13/11/Q17

A ripple tank is used to show wave behaviour. The dipper vibrates up and down at a constant frequency.

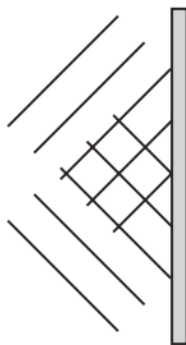


What happens to the frequency and to the speed of the wave as it reaches the glass block?

	frequency	speed
A	decreases	decreases
B	decreases	increases
C	remains the same	decreases
D	remains the same	increases

47. O/N/12/11/Q21

The diagram shows the pattern of waves in a ripple tank.

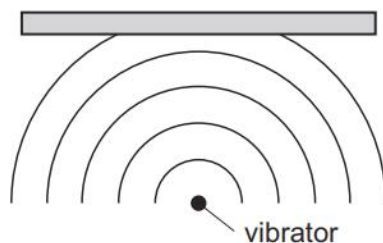


What does the pattern show?

- A** waves being reflected
- B** waves being refracted
- C** waves changing frequency
- D** waves changing speed

48. O/N/12/12/Q21

In a ripple tank, a vibrator produces circular wavefronts which hit a flat surface.



The reflected wavefronts are also parts of circles. Where is the centre of these circles?

A ●

B ●



D ●
vibrator

49. M/J/12/12/Q21

A wave has a frequency of 2 Hz.

How many waves are produced in one minute?

A 2×60

B $\frac{60}{2}$

C 2

D $\frac{2}{60}$

50. M/J/12/11/Q21

The speed of radio waves is c . A radio station transmits waves with a wavelength λ .

What is the frequency of the transmission?

A $\frac{c}{\lambda}$

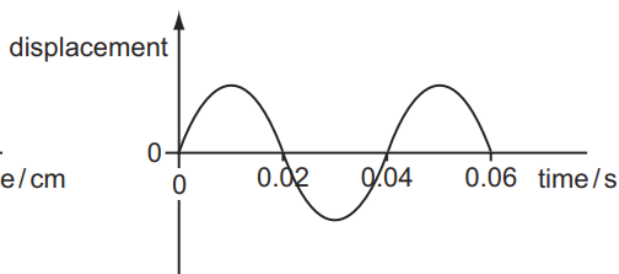
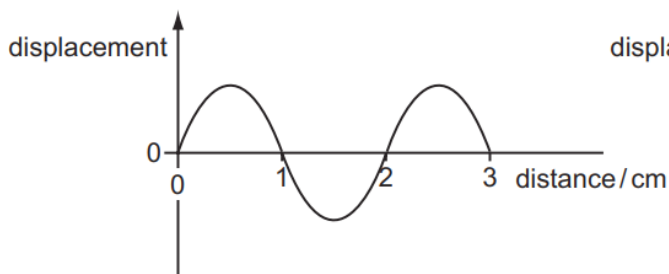
B $\frac{\lambda}{c}$

C $\frac{1}{\lambda}$

D $c\lambda$

51. M/J/11/12/Q22

The displacement-distance and displacement-time graphs are for a water wave in a ripple tank.



What is the speed of the water wave?

A 0.02 cm/s

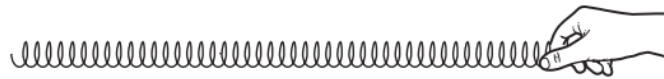
B 0.08 cm/s

C 25 cm/s

D 50 cm/s

52. O/N/11/11/Q23

A student uses a spring to demonstrate waves. He moves the spring with his hand.



spring placed on bench

Which diagram demonstrates the type of wave produced by a source of sound?

A



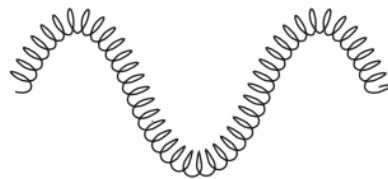
B



C



D



53. O/N/10/12/Q21

During a thunderstorm, there is an interval of 1.70 s between an observer seeing the lightning and hearing the thunder. The speed of sound is 340 m/s.

What is the distance between the observer and the storm?

- A** 100 m **B** 200 m **C** 578 m **D** 1160 m

54. M/J/10/12/Q20

A flash of lightning and the corresponding sound of the thunder are detected 6 s apart. A student calculates that the lightning struck about 1800 m away.

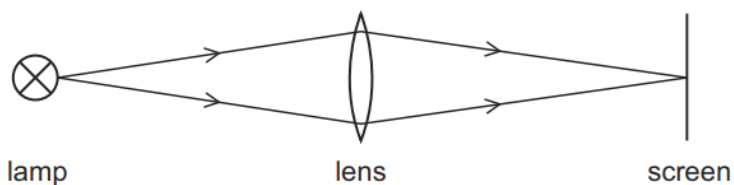
On which assumption is the calculation based?

- A** Light reaches us almost instantaneously, but sound travels at 300 m/s.
B Light travels 300 m/s faster than sound.
C Light travels 300 times faster than sound.
D The sound of the thunder was emitted 6 s after the flash.

55. M/J/10/12/Q19

Which diagram shows an example of a longitudinal wave?

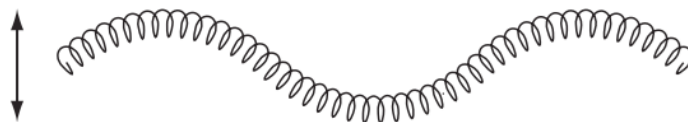
- A** light travelling from a lamp to a screen



- B** a spring pushed backwards and forwards



- C** a spring pushed up and down

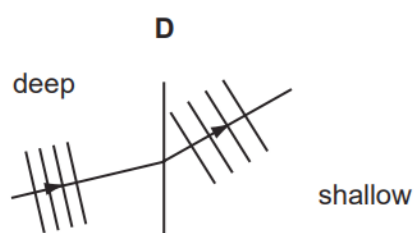
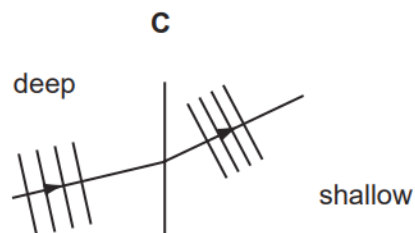
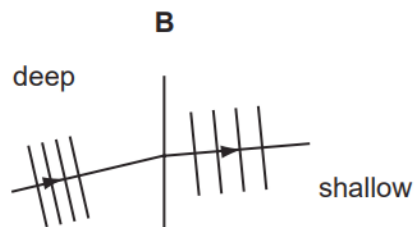
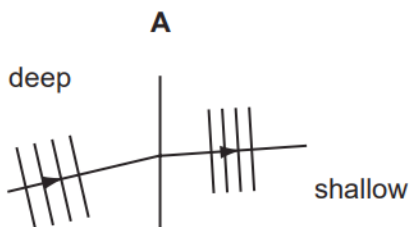


- D** a water ripple caused by a dipper moving up and down



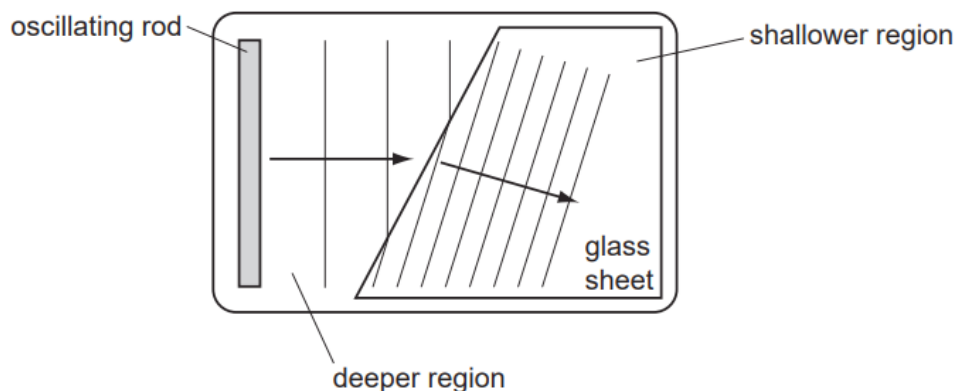
56. M/J/09/Q19

Which diagram correctly represents water waves travelling from deep water to shallow water?



57. O/N/08/Q22

The diagram shows the refraction of water waves in a ripple tank. The water is shallower above the glass sheet.

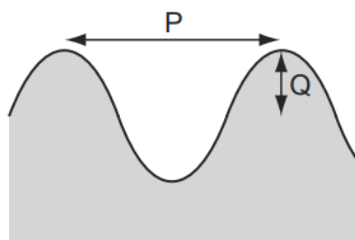


When crossing into the shallower region, what is the effect on the frequency and on the speed of the waves?

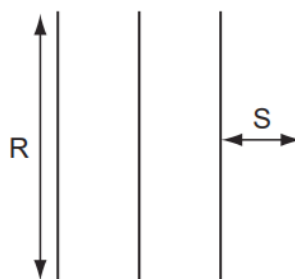
	wave frequency	wave speed
A	changes	changes
B	changes	unchanged
C	unchanged	changes
D	unchanged	unchanged

58. M/J/08/Q20

The diagrams show different views of a water wave in a ripple tank.



cross-section of wave



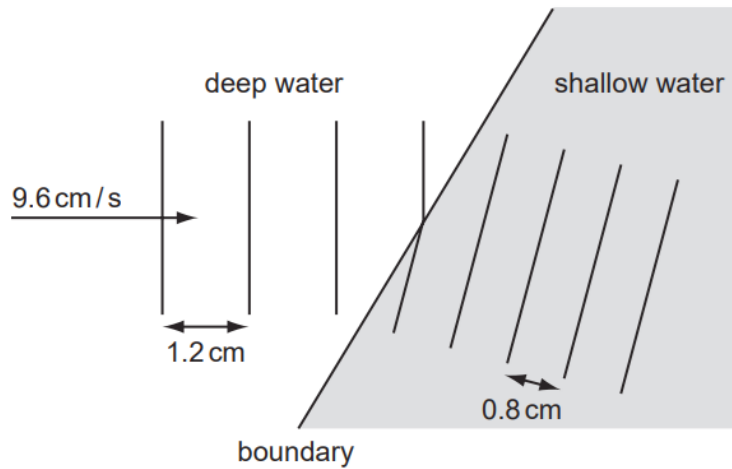
crests seen from above

Which letters represent a wavelength and a wavefront?

	wavelength	wavefront
A	P	R
B	P	S
C	Q	R
D	Q	S

59. M/J/08/Q21

A ripple tank is used to demonstrate refraction of plane water waves.



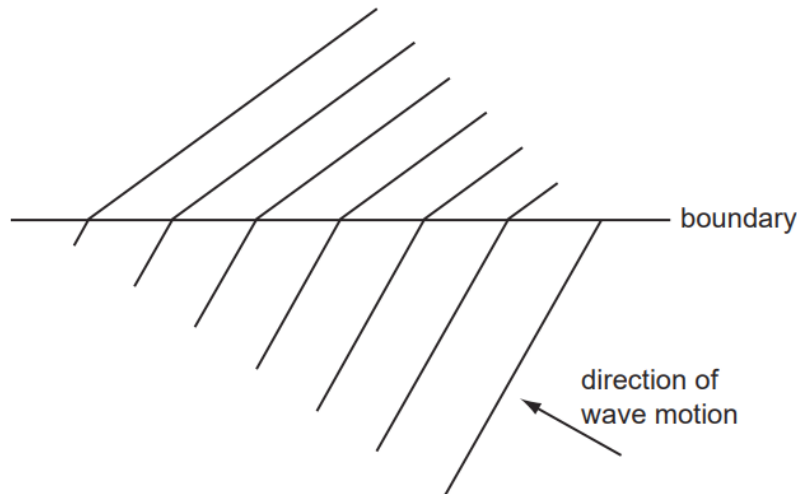
Waves in deep water have a wavelength of 1.2 cm and a speed of 9.6 cm/s. The wavelength of the waves in shallow water is 0.8 cm.

What is the speed of the waves in the shallow water?

- A** 6.4 cm/s **B** 8.0 cm/s **C** 9.6 cm/s **D** 14.4 cm/s

60. O/N/07/Q19

The diagram shows the refraction of water waves as they cross a boundary in a ripple tank.

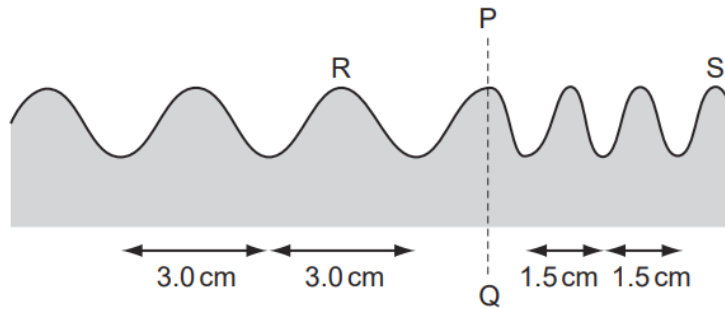


What causes this refraction?

- A** a change in frequency due to a change in depth
B a change in frequency due to a change in wavelength
C a change in speed due to a change in depth
D a change in speed due to a change in frequency

61. M/J/07/Q19

The diagram shows a water wave in a ripple tank.



The wave has a speed of 12 cm/s at R.

The wave crosses a boundary PQ where the distance between crests changes from 3.0 cm to 1.5 cm.

What is the speed of the wave at S?

- A** 3.0 cm/s **B** 6.0 cm/s **C** 12 cm/s **D** 24 cm/s

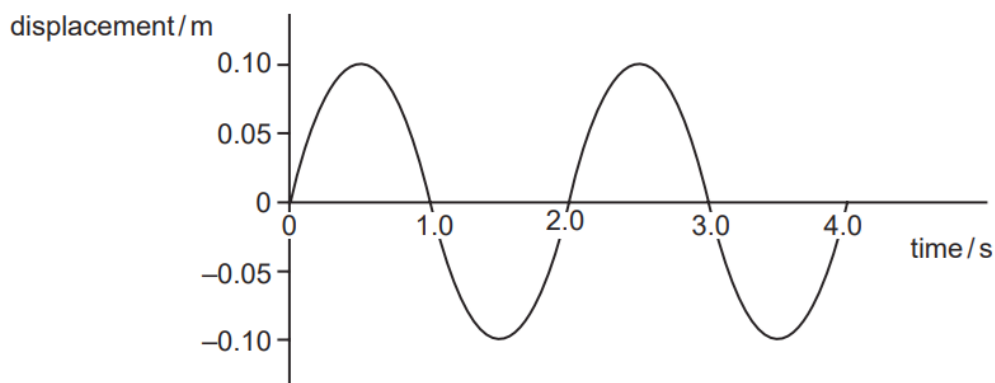
62. O/N/06/Q19

What is meant by the term *wavefront*?

- A** the distance between successive peaks of a wave
B the distance between the trough and the peak of a wave
C a line joining points along the peak of a wave
D a line joining the trough and the peak of a wave

63. O/N/06/Q20

The diagram shows how displacement varies with time as a wave passes a fixed point.

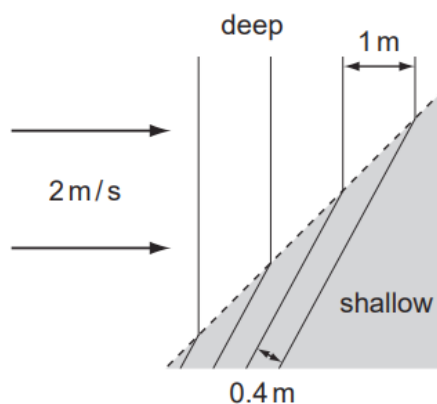


What is the frequency of this wave?

- A** 0.25 Hz **B** 0.50 Hz **C** 1.0 Hz **D** 2.0 Hz

64. M/J/06/Q22

Waves pass from deep water to shallow water and refraction occurs.

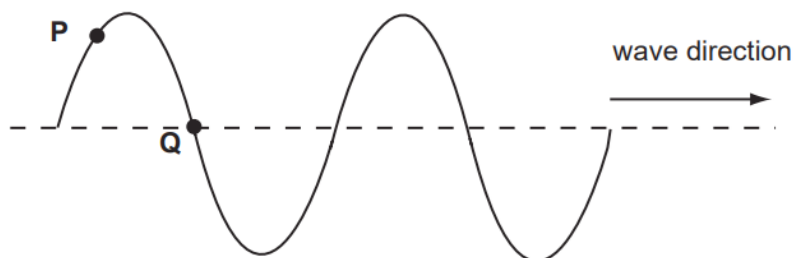


What is the speed of the waves in the shallow water?

- A** 0.2 m/s **B** 0.8 m/s **C** 2.0 m/s **D** 5.0 m/s

65. O/N/05/Q20

The diagram shows a wave on a string with two points **P** and **Q** marked. The wave is moving in the direction shown.



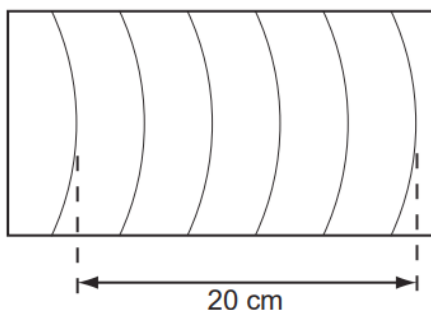
What will happen next?

- A** **P** will move to the right.
B **P** will move up.
C **Q** will not move.
D **Q** will move up.

66. O/N/05/Q21

The dipper in a ripple tank vibrates at a frequency of 4.0 Hz and the resulting wave pattern is photographed.

The distance between the two crests shown is 20 cm.

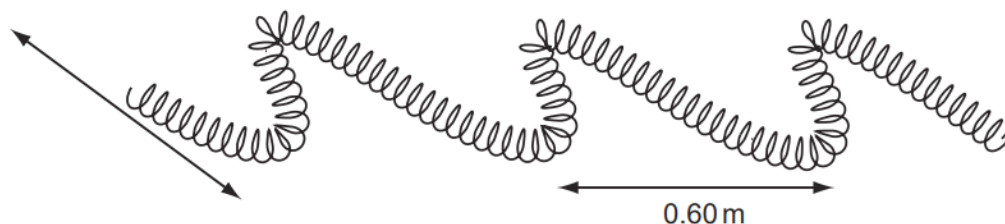


What is the speed of the wave?

- A** 4 cm/s **B** 5 cm/s **C** 16 cm/s **D** 20 cm/s

67. M/J/05/Q20

The diagram shows part of a spring that is shaken from side to side to produce a wave.



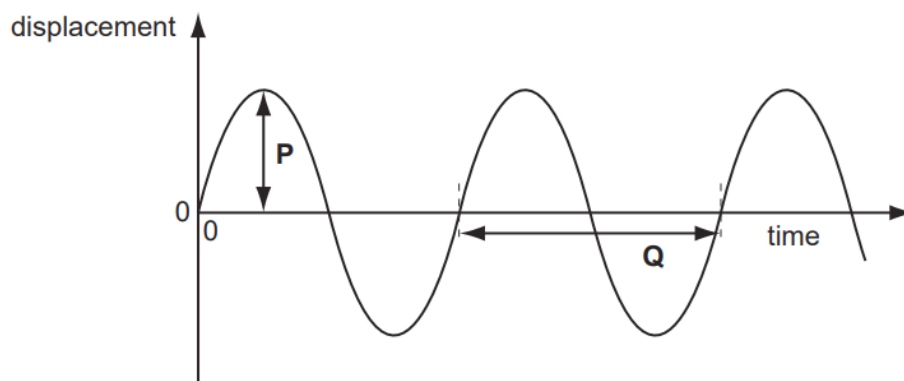
The distance between successive peaks is 0.60 m and the frequency is 2.5 Hz.

How long does it take for a wave to travel 3.0 m along the spring?

- A** 0.20 s **B** 0.50 s **C** 2.0 s **D** 5.0 s

68. O/N/04/Q22

The diagram shows a graph of wave motion.

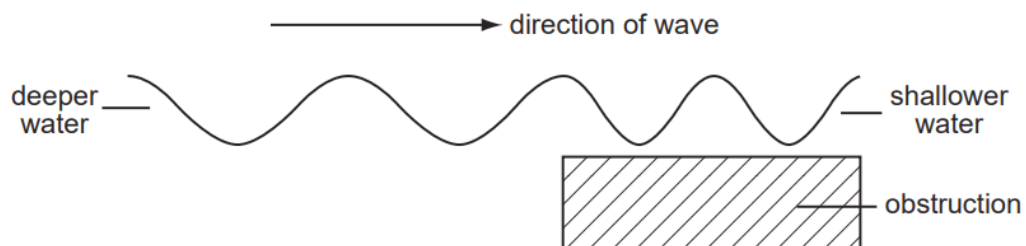


Which quantities are shown by distances **P** and **Q**?

	P	Q
A	amplitude	period
B	amplitude	wavelength
C	half the amplitude	period
D	half the amplitude	wavelength

69. M/J/04/Q19

The diagram shows a wave moving into shallower water.



The wavelength of the waves is reduced because

- A** both the frequency and the speed decrease.
- B** both the frequency and the speed increase.
- C** only the frequency increases.
- D** only the speed decreases.